A First Course in Abstract Algebra

Section 7: Generating Sets and Cayley Digraphs

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

Contents

7 Generating Sets and Cayley Digraphs 3

Exercise 7.1: Two and three modulo twelve 3

Exercise 7.2: Four and six modulo twelve 3

Exercise 7.3: Eight and ten modulo eighteen 3

Exercise 7.4: Twelve and thirty modulo thirty-six 3

Exercise 7.5: Twelve and forty-two in the integers 3

Exercise 7.6: Three integer generators 3

Exercise 7.7: Products in the eight-element digraph 3

Exercise 7.8: The square digraph table 4

Exercise 7.9: The alternating-edge hexagon table 4

Exercise 7.10: The two-triangle digraph table 4

Exercise 7.11: Recognizing commutativity from paths 5

Exercise 7.12: The eight-element example is nonabelian 5

Exercise 7.13: Recognizing cyclicity 5

Exercise 7.14: The inner triangle is not a subgroup 5

Exercise 7.15: A Cayley digraph with generators one and two 6

Exercise 7.16: A Cayley digraph with generators two and five 6
Exercise 7.17: Relations read from closed paths 7
Exercise 7.18: Minimal digraphs at order four 7
Exercise 7.19: Two involutions generating a nonabelian group of order twice n 7

7 Generating Sets and Cayley Digraphs

Exercise 7.1: Two and three modulo twelve

The generated subgroup contains $3-2=1$, because subgroups are closed under addition and negatives. Once it contains 1, it contains every residue. Therefore it is $\mathbb{Z}_{12} = \{0,1,2,3,4,5,6,7,8,9,10,11\}$.

Exercise 7.2: Four and six modulo twelve

Every integer combination of 4 and 6 is even, so the subgroup lies inside the even residues. It contains $6-4=2$, and hence every even residue. Thus it is exactly $\{0,2,4,6,8,10\}$.

Exercise 7.3: Eight and ten modulo eighteen

Every combination of $8, 10$ and the modulus 18 is even, so no odd residue can occur. The difference $10-8=2$ belongs, and its multiples give all even residues. Thus the subgroup is $\{0,2,4,6,8,10,12,14,16\}$.

Exercise 7.4: Twelve and thirty modulo thirty-six

The difference $30-2\cdot 12=6$ belongs. Every input and the modulus are divisible by 6, so only multiples of 6 occur, and all of them occur by adding 6. The subgroup is therefore $\{0,6,12,18,24,30\}$.

Exercise 7.5: Twelve and forty-two in the integers

All generated elements are combinations $12m+42n$, hence multiples of 6. Conversely $42-3\cdot 12=6$ belongs, so every integer multiple of 6 belongs. The subgroup is $6\mathbb{Z} = \{\ldots,-12,-6,0,6,12,\ldots\}$.

Exercise 7.6: Three integer generators

All three inputs $18, 24, 39$ are divisible by 3. The generated subgroup contains $24-18=6$ and then $39-6\cdot 6=3$. It therefore contains every multiple of 3, and contains no other integer. The answer is $3\mathbb{Z} = \{\ldots,-6,-3,0,3,6,\ldots\}$.

Exercise 7.7: Products in the eight-element digraph

The outer solid cycle gives $a^4=e$. Dashed edges give $b^2=e$, and the reversed inner solid cycle gives $ba=a^{-1}b$. Thus $ba^k=a^{-k}b$ for every integer k.

(a) $(a^2b)a^3 = a^2a^{-3}b = a^{-1}b = a^3b$.

(b) $(ab)(a^3b) = aa^{-3}b^2 = a^{-2} = a^2$.

(c) $b(a^2b) = a^{-2}b^2 = a^{-2} = a^2$.

Each result can also be checked by starting at the first factor's vertex and following the right factor's directed edge sequence.

Exercise 7.8: The square digraph table

Both indicated generator types are involutions, and traversing one of each takes e to c in either order. Thus $a^2 = b^2 = e$, $ab = ba = c$, giving

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

For an order-two generator an undirected edge stands for the two opposite directed edges. The resulting group is the Klein four-group.

Exercise 7.9: The alternating-edge hexagon table

Read the two edge types as involutions a, b. The paths give $c = ab$, $d = ba$, and $f = aba = bab$. One concrete model is $a = (12)$ and $b = (23)$, so $c = (123)$, $d = (132)$, and $f = (13)$. Multiplying these permutations, with the rightmost one applied first, yields the table in the requested alphabetical order:

	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b	f	d
b	b	d	e	f	a	c
c	c	f	a	d	e	b
d	d	b	f	e	c	a
f	f	c	d	a	b	e

For example $ab = c$ but $ba = d$, so this six-element group is nonabelian. The model also verifies the whole table's associativity.

Exercise 7.10: The two-triangle digraph table

The outside arrows give $a^3 = e$ and $c = a^2$. The dashed edges give $b^2 = e$, $ab = f$, $a^2b = d$. Reading the inside arrows yields $ba = a^{-1}b$, so this is the triangular dihedral group. Using $a = (123)$, $b = (12)$ realizes these labels concretely and gives

	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	c	f	e	b	d
b	b	d	e	f	a	c
c	c	e	d	a	f	b
d	d	f	c	b	e	a
f	f	b	a	d	c	e

For example $ab = f$ and $ba = d$, consistent with the distinct paths in the figure. This diagram and the preceding one describe isomorphic groups, but their element names represent different permutations.

Exercise 7.11: Recognizing commutativity from paths

Compare the endpoints of paths using two generator types in opposite orders. If some s-then-t path ends differently from the corresponding t-then-s path at the same starting point, the group is noncommutative. Conversely, if every pair of generators commutes, their inverses also commute; any two words in generators and inverses can then be reordered past each other. Since such words are all group elements, the group is abelian. Thus this two-step-path comparison is a complete criterion.

Exercise 7.12: The eight-element example is nonabelian

Starting at e, traversing an a edge and then a b edge ends at ab, whereas traversing b then a ends at a^3b. These are different labeled vertices in Fig. 7.11(b). Therefore $ab \neq ba$ and the group is noncommutative.

Exercise 7.13: Recognizing cyclicity

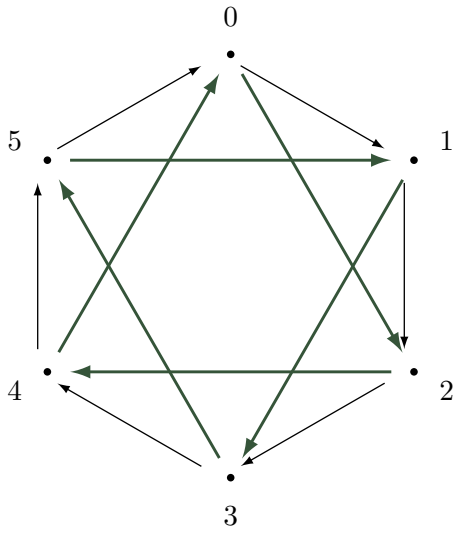
A single generator type forming a directed cycle through every vertex proves cyclicity, but its absence does not disprove cyclicity. A diagram might use a redundant generating set without displaying any individual generator of the whole group. For example $\mathbb{Z}_6$ generated by $\{2, 3\}$ is cyclic, while the 2-edges form triangles and the 3-edges form pairs. Thus cyclicity need not be visually apparent from one edge type; one may need a repeated word in several edge types.

Exercise 7.14: The inner triangle is not a subgroup

The smaller triangle in Fig. 7.9(b) has the vertices $\{1, 3, 5\}$. It does not contain the additive identity 0, so it cannot be a subgroup of $\mathbb{Z}_6$. It is also not closed: $1 + 1 = 2$ is outside it. It is a translated copy of the subgroup $\{0, 2, 4\}$, which explains the identical triangle shape without making it a subgroup.

Exercise 7.15: A Cayley digraph with generators one and two

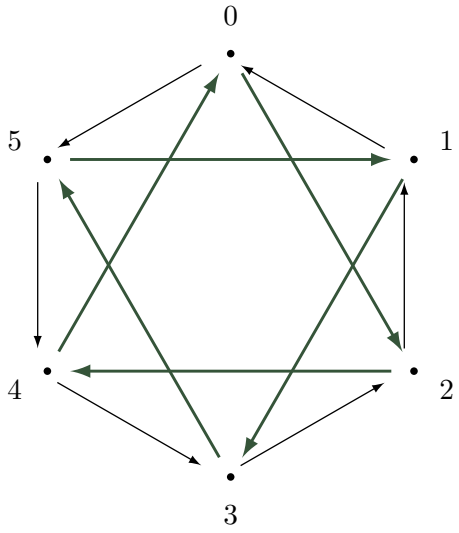
Place the six residues around a hexagon in the order $0, 1, 2, 3, 4, 5$ clockwise. Draw a thin black arrow $x \to x + 1$ and a thick green arrow $x \to x + 2$, reducing modulo 6:



The black arrows form $0 \to 1 \to 2 \to 3 \to 4 \to 5 \to 0$. The green arrows form $0 \to 2 \to 4 \to 0$ and $1 \to 3 \to 5 \to 1$. Thus each vertex has exactly one incoming and one outgoing arrow of each generator type, as required.

Exercise 7.16: A Cayley digraph with generators two and five

Use the same labeled hexagon. Draw thick green arrows $x \to x+2$ and thin black arrows $x \to x+5 = x - 1$ modulo 6:



The green triangles are $0 \to 2 \to 4 \to 0$ and $1 \to 3 \to 5 \to 1$. The black outside cycle is $0 \to 5 \to 4 \to 3 \to 2 \to 1 \to 0$. Since 5 alone is a generator, the combined diagram is connected and represents all of $\mathbb{Z}_6$.

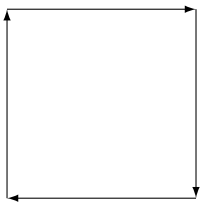
Exercise 7.17: Relations read from closed paths

(a) Begin at the identity and follow a closed path. Write down the generator for each forward edge and its inverse for each backward edge. The resulting word is a product equal to e, hence a relation. Conversely any such relation returns the identity to itself when its word is followed.

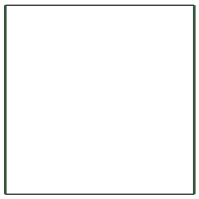
(b) In Fig. 7.11(b), the outside four-cycle gives $a^4 = e$, a dashed round trip gives $b^2 = e$, and the path expressing $ba = a^{-1}b$ gives $baba = e$. Thus three relations are $a^4 = e$, $b^2 = e$, and $(ba)^2 = e$. The last can also be written $bab = a^{-1}$.

Exercise 7.18: Minimal digraphs at order four

For the cyclic group use one generator: a directed four-cycle. For the Klein group one generator never suffices, since every element has order at most 2; use two involutions and a square with different edge types in the two directions. Here a double-headed edge is represented by an undirected line, as usual for an involution.



$\mathbb{Z}_4$: one generator



V: two generators

Each square has four vertices at its corners. In the right-hand diagram horizontal edges have one generator type and vertical edges the other; both types are their own reverses.

Exercise 7.19: Two involutions generating a nonabelian group of order twice n

Take symmetries of a regular n-gon, $n \geq 3$. Let r rotate through $2\pi/n$ and s reflect in an axis through its center. The symmetries are exactly r^k and $r^k s$ for $0 \leq k < n$: these $2n$ transformations are distinct because the first preserve orientation and the second reverse it, and the rotations themselves are distinct. They are closed under composition because $sr = r^{-1}s$.

Choose two order-two generators.

The elements s and rs are reflections. They satisfy $s^2 = e$ and $(rs)^2 = r(sr)s = rr^{-1}s^2 = e$, and neither is the identity. Their product $(rs)s = r$ gives the basic rotation, so they generate every $r^k, r^k s$.

Check noncommutativity.

The products rs and $sr = r^{-1}s$ differ: equality would give $r = r^{-1}$, contradicting that r has order $n \geq 3$. Thus the group is nonabelian, has exactly $2n$ elements, and is generated by two elements of order 2.